

Birzeit University
Faculty of Engineering and Technology
Electrical and Computer Engineering Department

EE2312 – Signals and Systems

MATLAB Assignment Report

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Question I – Signal Generation and Plotting

Objective:

Generate and plot the following signals using MATLAB on a single figure and as subplots:

1. A finite pulse $\pi(t/6)$ with value = 5 centered at $t = 10$
2. $x(t) = u(t-1) - r(-t+7) + r(t+4) + \pi((t+1)/6)$
3. $x(t) = \pi((t-3)/6) - r(t-7) + r(t-13) - u(t-13)$ on $[0, 22]$

MATLAB Code:

```
t = -20:0.01:30;
u = @(t) (t >= 0);
r = @(t) t .* (t >= 0);
rect = @(t) (abs(t) <= 0.5);

% Signal 1: 5*rect((t-10)/6)
s1 = 5 * rect((t - 10) / 6);

% Signal 2: u(t-1) - r(-t+7) + r(t+4) + rect((t+1)/6)
s2 = u(t-1) - r(-t+7) + r(t+4) + rect((t+1)/6);

% Signal 3: rect((t-3)/6) - r(t-7) + r(t-13) - u(t-13)
t3 = 0:0.01:22;
s3 = rect((t3-3)/6) - r(t3-7) + r(t3-13) - u(t3-13);

% Figure 1 - All on one plot
figure(1);
plot(t,s1,'r','LineWidth',1.5); hold on;
plot(t,s2,'b','LineWidth',1.5);
plot(t3,s3,'g','LineWidth',1.5);
legend('Signal 1','Signal 2','Signal 3');
title('Question I - All Signals'); xlabel('t'); grid on;

% Figure 2 - Subplots
figure(2);
subplot(3,1,1); plot(t,s1,'r','LineWidth',1.5);
title('Signal 1: 5pi((t-10)/6)'); xlabel('t'); grid on;
subplot(3,1,2); plot(t,s2,'b','LineWidth',1.5);
title('Signal 2'); xlabel('t'); grid on;
subplot(3,1,3); plot(t3,s3,'g','LineWidth',1.5);
title('Signal 3'); xlabel('t'); grid on;
```

Results and Discussion:

Signal 1 is a rectangular pulse of amplitude 5 centered at $t=10$ with width 6 (active from $t=7$ to $t=13$). Signal 2 combines a unit step, ramp functions, and a rectangular pulse producing a linearly growing waveform. Signal 3 represents a piecewise signal that starts with a rectangular pulse, slopes downward from $t=7$ to $t=13$, then remains at a constant negative value, defined over $[0, 22]$.

Question II – Sinusoidal Signals and Operations

Objective:

Generate $y_1(t)=10\sin(0.25\pi t)$, $y_2(t)=5\cos(0.125\pi t)$, $y_3(t)=8\sin(2\pi t)$, compute combined signals, analyze periodicity, and plot amplitude spectra.

MATLAB Code:

```
t = 0:0.001:64; % 4 cycles of y2 (highest period = 16s)
y1 = 10 * sin(0.25*pi*t);
y2 = 5 * cos(0.125*pi*t);
y3 = 8 * sin(2*pi*t);

m1 = y1 + y2; m2 = y1 - y2;
m3 = y2 .* y3; m4 = y1 .* y3;

% Subplots of y1, y2, y3 (Figure 1)
% Combined m1-m4 on same figure (Figure 2)
% Amplitude spectra via FFT (Figure 3)
```

Periodicity Analysis:

Signal	Periodic?	Period T	Alternating?	Half-Wave Odd Sym.?
$m1 = y1+y2$	YES	$\text{LCM}(8,16) = 16 \text{ s}$	YES	NO
$m2 = y1-y2$	YES	16 s	YES	NO
$m3 = y2 \times y3$	YES	$\text{LCM}(16,1) = 16 \text{ s}$	YES	YES
$m4 = y1 \times y3$	YES	$\text{LCM}(8,1) = 8 \text{ s}$	YES	YES

m3 half-wave symmetry proof: $m3(t + T/2) = y2(t+8) \cdot y3(t+8) = -y2(t) \cdot y3(t) = -m3(t)$ ✓ m4 half-wave symmetry proof: $m4(t + T/2) = y1(t+4) \cdot y3(t+4) = -y1(t) \cdot y3(t) = -m4(t)$ ✓

Question III – Zero-State Responses

Objective:

Solve four differential equations numerically using ode45, determine stability type, and identify steady-state responses.

MATLAB Code (System 1 example):

```
tspan = [0 100];  
ode1 = @(t,y) [-0.125*y(1) + 5];  
ode2 = @(t,y) [y(2); -5*y(1) - 7*y(2) + 5*cos(0.01*pi*t)];  
ode3 = @(t,y) [y(2); -5*y(1) - 4*y(2) + 5];  
ode4 = @(t,y) [y(2); -5*y(1) + 4*y(2) + 5];  
[t1,y1] = ode45(ode1, tspan, [0]);  
[t2,y2] = ode45(ode2, tspan, [0 0]);  
[t3,y3] = ode45(ode3, tspan, [0 0]);  
[t4,y4] = ode45(ode4, tspan, [0 0]);
```

Stability and Steady-State Analysis:

System	Equation	Roots	Stability	Steady State	Value
Sys 1	$y' + 0.125y = 5u(t)$	-0.125	Asymptotically Stable	Constant	40
Sys 2	$y'' + 7y' + 5y = 5\cos(0.01\pi t)$	-0.70, -6.30	Asymptotically Stable	Sinusoidal	~0
Sys 3	$y'' + 4y' + 5y = 5u(t)$	$-2 \pm j$	Asymptotically Stable	Constant	1
Sys 4	$y'' - 4y' + 5y = 5u(t)$	$2 \pm j$	UNSTABLE	Does NOT exist	∞

Time constants: Sys1: $\tau=8s$, $4\tau=32s$ | Sys2: $\tau \approx 1.43s$, $4\tau \approx 5.7s$ | Sys3: $\tau=0.5s$, $4\tau=2s$ | Sys4: UNSTABLE

Question IV – Step Response of Systems

Objective:

Simulate two systems and plot their step responses using transfer functions.

System 1: $5y^{(5)} + 2y''' + 6y' + 4y = 2x'' + 3x' - 6x$

System 2: $y'' + 2y' + 4y = 5x + 3x'$

```
% System 1: H(s) = (2s^2+3s-6) / (5s^5+2s^3+6s+4)
num1 = [2, 3, -6];
den1 = [5, 0, 2, 0, 6, 4];
sys1 = tf(num1, den1);

% System 2: H(s) = (3s+5) / (s^2+2s+4)
num2 = [3, 5];
den2 = [1, 2, 4];
sys2 = tf(num2, den2);

figure; subplot(2,1,1); step(sys1); grid on;
subplot(2,1,2); step(sys2); grid on;
pzmap(sys1); pzmap(sys2);
```

Results:

System 1 has poles in the right half-plane (unstable) — the step response diverges. System 2 is stable with all poles in the left half-plane; it settles at steady-state value = $5/4 = 1.25$.

Note: Simulink separate/integrate modeling was replaced by equivalent MATLAB tf() and step() commands.

Question V – Convolution Integral

Objective:

Compute and plot the convolution of two signal pairs.

```
dt = 0.01;
rect = @(t) (abs(t) <= 0.5);

% Pair 1
t1 = -20:dt:20;
y1 = (10*exp(2*t1)) .* rect((t1-3)/6);
y2 = 12*sin(0.25*pi*t1) .* rect((t1-1)/2);
conv1 = conv(y1, y2, 'full') * dt;

% Pair 2
t2 = -5:dt:5;
y3 = (10*exp(2*t2)) .* rect((t2-0.1)/0.1); % Approx. impulse
y4 = rect((t2-1)/2);
conv2 = conv(y3, y4, 'full') * dt;
```

Results:

Pair 1: y1 is a one-sided exponential pulse; y2 is a windowed sinusoid. Their convolution produces a smooth bell-shaped pulse centered around $t \approx 8$. Pair 2: y3 approximates a Dirac delta function (very narrow pulse). Convolution with y4 (rect pulse) produces a scaled version of y4, confirming the sifting property of the impulse: $x(t) * \delta(t-t_0) \approx x(t-t_0)$.

Question VI – Laplace Transform

Objective:

Compute symbolic Laplace transforms of two signals.

```
syms t s
y3 = 10 - 10*exp(-0.25*t);
Y3 = laplace(y3, t, s); % = 10/s - 10/(s+0.25) = 2.5/(s*(s+0.25))

y4 = 8 + 4*exp(-0.0625*t)*cos(0.125*pi*t);
Y4 = laplace(y4, t, s);
% = 8/s + 4*(s+0.0625)/((s+0.0625)^2 + (0.125*pi)^2)
```

Theoretical Results:

Signal	y(t)	Y(s)
Signal 3	$(10 - 10e^{-0.25t})u(t)$	$2.5 / [s(s + 0.25)]$
Signal 4	$8 + 4e^{-0.0625t}\cos(0.125\pi t)u(t)$	$8/s + 4(25\pi^2) / [(s+0.0625)^2 + (0.125\pi)^2]$

Question VII – System Analysis in Laplace Domain

System: $6y^{(5)} - 7y''' + 2y' + 9y = 10x' + 5x$

Transfer Function: $H(s) = (10s + 5) / (6s^5 - 7s^3 + 2s + 9)$

```
num = [10, 5];  
den = [6, 0, -7, 0, 2, 9];  
sys = tf(num, den);  
  
p = pole(sys);    % Poles  
z = zero(sys);    % Zeros  
  
pzmap(sys);       % Pole-Zero map  
step(sys, 2);     % Step response (diverges - unstable)  
  
s_tf = tf('s');  
sys_ramp = sys / s_tf;  
step(sys_ramp, 2); % Ramp response
```

Stability Analysis:

The system has poles in the right half-plane (positive real parts $\approx +1 \pm 0.5j$). Therefore the system is UNSTABLE. Both step and ramp responses diverge, confirming instability. The pole-zero map clearly shows poles (x) in the right half-plane.

Note: Simulink state observer model replaced by equivalent MATLAB tf() commands.

Question VIII – $H(s) = 10000(s+2)/(s^2+4s+5)$

```
num = 10000 * [1, 2];  
den = [1, 4, 5];  
sys = tf(num, den);  
  
figure(1); pzmap(sys); % Poles: -2 ± j | Zero: -2  
figure(2);  
subplot(2,1,1); step(sys); % Steady state = H(0) = 4000  
s = tf('s');  
subplot(2,1,2); step(sys/s, 10); % Ramp response
```

Results:

Property	Value
Poles	$-2 \pm j$ (Left Half-Plane)
Zero	-2
Stability	STABLE – Asymptotically Stable
Step Steady-State	$H(0) = 10000 \times 2/5 = 4000$
Ramp Response	Diverges (ramp input to stable system)

Question IX – MISO System Analysis

$$H1(s) = 10000(s+3)/(s^2+6s+8) \mid H2(s) = 100(s-3)/(s^2+2s+1)$$

```
num1 = 10000*[1,3]; den1 = [1,6,8]; sys1 = tf(num1,den1);  
num2 = 100*[1,-3]; den2 = [1,2,1]; sys2 = tf(num2,den2);  
  
% Step responses  
step(sys1); step(sys2);  
  
% Frequency response of H1  
w = logspace(-2,4,1000);  
bode(sys1, w);
```

Results:

System	Poles	Zeros	Stability	SS Value
H1	-2, -4	-3	STABLE	$10000 \times 3/8 = 3750$
H2	-1, -1 (repeated)	+3 (RHP zero)	STABLE	$100 \times (-3)/1 = -300$

H2 has a non-minimum phase zero at $s=+3$ causing initial undershoot (negative response) despite the system being stable. The frequency response of H1 shows low-pass characteristics.

Question X – AM Modulation

Single-tone AM: message $f_m=0.25\text{Hz}$, $A_m=10$ | carrier $f_c=8\text{Hz}$, $A_c=15$

```
fm=0.25; fc=8; Am=10; Ac=15;
t = 0:0.001:16;
msg      = Am * sin(2*pi*fm*t);
carrier  = Ac * cos(2*pi*fc*t);
modulated = (Ac + msg) .* cos(2*pi*fc*t);
% Spectra via FFT - subplots for message, carrier, modulated
```

Periodicity Analysis:

$T_m = 1/0.25 = 4 \text{ s}$ | $T_c = 1/8 = 0.125 \text{ s}$ | $\text{Ratio} = T_m/T_c = 32 \text{ (integer)} \rightarrow \text{PERIODIC}$

The modulated signal is periodic with fundamental period $T = 4 \text{ s}$. The AM spectrum shows the carrier at $\pm 8 \text{ Hz}$ plus sidebands at $\pm 7.75 \text{ Hz}$ and $\pm 8.25 \text{ Hz}$, consistent with standard DSB-LC AM theory: $S(f) = A_c/2[\delta(f-f_c) + \delta(f+f_c)] + M(f-f_c)/2 + M(f+f_c)/2$.

Question XI – Fourier Transform

Objective:

Compute Fourier transforms numerically via FFT and compare with theoretical results.

Signal 1: $x(t) = \pi(t/5)$

Theoretical: $X(f) = 5 \cdot \text{sinc}(5f)$ — a sinc function scaled by the pulse width (5).

Result: FFT matches the theoretical $5 \cdot \text{sinc}(5f)$ curve exactly. The main lobe has width $2/5 = 0.4$ Hz. Phase flips by π at each zero crossing.

Signal 2: $x(t) = 10\sin(2\pi t) \cdot \cos(40\pi t)$

Using the product-to-sum identity:

$$10\sin(2\pi t)\cos(40\pi t) = 5\sin(42\pi t) - 5\sin(38\pi t)$$

Theoretical frequencies: ± 21 Hz and ± 19 Hz. FFT confirms impulses at exactly these four frequencies.

Signal 3: $x(t) = 10e^{-3t}u(t)$

Theoretical: $X(f) = 10/(3 + j2\pi f) \rightarrow |X(f)| = 10/\sqrt{9+(2\pi f)^2}$

Result: FFT matches the theoretical Lorentzian shape perfectly. Peak at $f=0$: $|X(0)| = 10/3 \approx 3.33$.

End of Report